

Contract

IQ

- **Team Name :**
 - Team_404
- **University :**
 - University of Moratuwa
- **Target SBU :**
 - Centre of Excellence (COE) - Hemas Holdings PLC
- **Target Function:**
 - Legal & Contract Governance / Procurement Operations
- **Challenge track:**
 - D2D (Day to Day) Corporate Use Cases | AI Powered Automation

IMPACT STATEMENT

- The Hemas Centre of Excellence (COE) manages thousands of contracts across six Strategic Business Units (pharmaceuticals, healthcare, manufacturing, logistics, consumer goods). Manual review by senior professionals currently takes 3-5 days per contract, creating a bottleneck.
- ContractIQ replaces this with a two-agent adversarial AI system: An AUDITOR agent identifies risks, which an ATTACKER agent actively challenges. The AUDITOR then produces a revised, adversarially-validated risk score, flagging documents for human escalation.
- The system uses a RAG knowledge base of Sri Lankan legal policies (PDPA 2022, Drugs Act, Companies Act) and Hemas's past annotations, ensuring jurisdiction-aware and continuously improving audits.
- COE expects an 99% reduction in manual review time, standardized risk assessment across all 6 SBUs, and PDPA compliance via local NLP anonymization. Estimated annual savings are LKR 4.2M+ in senior legal capacity.

PROBLEM IDENTIFICATION & BUSINESS CONTEXT

1. The Friction Point

- Manual contract review across 6 SBUs: 3–5 days/doc, single-reviewer perspective, no adversarial challenge mechanism, inconsistent standards, and bottlenecks that delay procurement and partnership decisions group-wide.

2. Stakeholders

- COE Legal & Procurement teams (primary). All 6 SBU GMs who depend on contract turnaround. CFO office managing financial risk exposure across the Hemas Group.

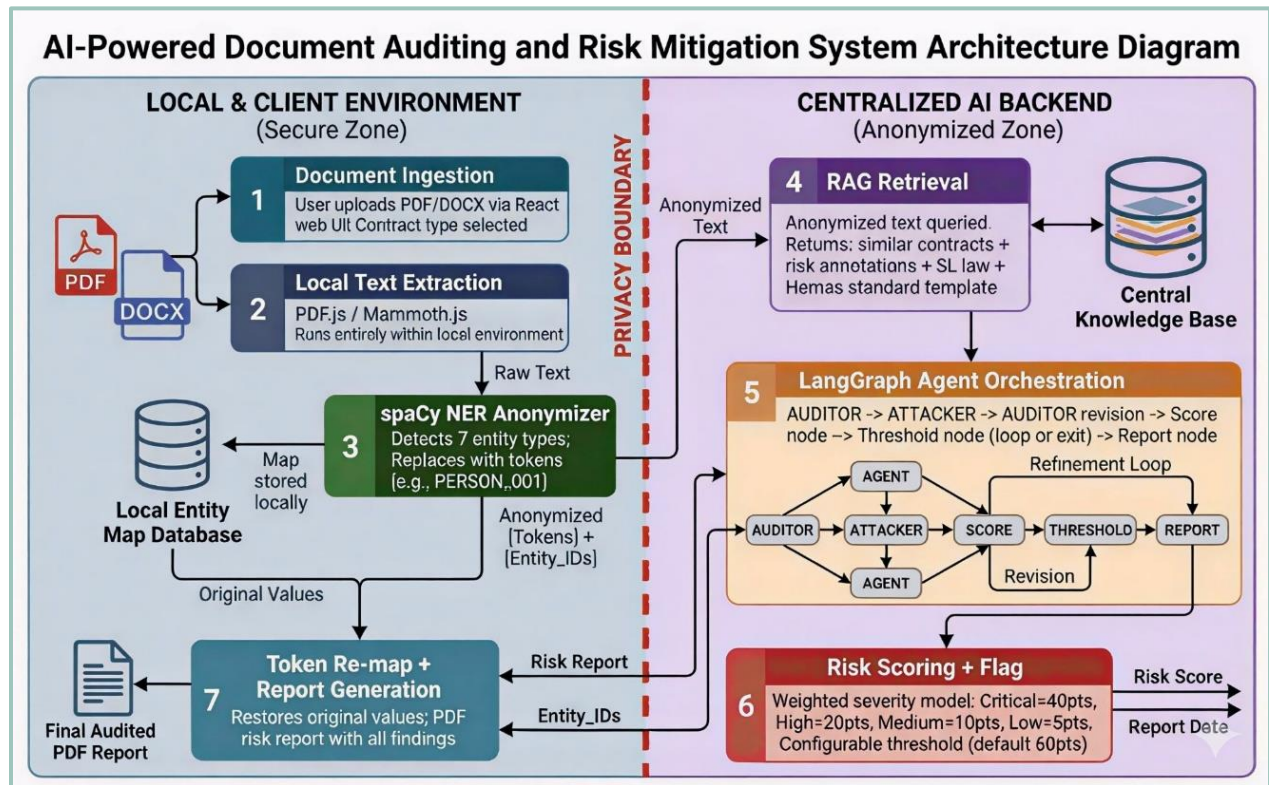
3. Why Critical in 2026

- Sri Lanka's PDPA 2022 enforcement is active. Hemas's multinational pharma partnerships are expanding. A missed liability clause in one agreement can cascade across SBUs. Manual review at current volume is no longer sustainable.
- Manual contract review undermines the COE's mandate: standardization, compliance, efficiency, and strategic alignment. Senior legal time is wasted on routine checks instead of strategic governance. ContractIQ uses AI to operationalize the COE's mandate.

AI METHODOLOGY

- **Logic Framework**
 - Combines local NLP for anonymization, RAG for Hemas-specific grounding, a LangGraph-orchestrated Multi-Agent Adversarial Loop, and a weighted Risk Scoring Engine.
- **Adversarial Audit Loop**
 - **Auditor Agent:** Extracts clauses and flags risks against Sri Lankan law (PDPA 2022, Companies Act) and Hemas templates.
 - **Attacker Agent:** Probes Auditor findings for gaps, challenges severity ratings, and identifies missed adversarial interpretations.
 - **Risk Scorer:** Aggregates findings via a weighted model; high-risk scores trigger human escalation, while lower scores trigger further Attacker passes.
- **RAG Knowledge Base**
 - Grounded in the Sri Lanka Legal Corpus, annotated historical Hemas contracts, and COE-approved standard templates.
- **The "ContractIQ" Edge**
 - **Adversarial Security:** Unlike single-pass LLMs, the Attacker actively hunts for what the Auditor missed.
 - **Privacy Boundary:** Local spaCy NER ensures no PII or sensitive commercial data leaves the internal environment.
 - **Continuous Learning:** The system improves over time as the knowledge base grows with every reviewed contract.
- **Data Requirements**
 - Publicly available Sri Lankan legislation
 - Annotated Historical Contracts from Hemas

TECHNICAL ARCHITECTURE



- **Tech Stack**

Frontend/ Backend	React.js, FastAPI, Python.
AI/LLM	LangGraph, Groq API (Llama 3.3 70B), ChromaDB
Local Processing	PDF.js, spaCy (NER), sentence-transformers.

- **Scalability**

- The pipeline is horizontally scalable, allowing multiple LangGraph instances to process contracts simultaneously. The RAG knowledge base expands automatically. For production, the Groq API can be replaced with an on-premise, open-source LLM (Llama 3, Mistral) hosted within Hemas infrastructure, eliminating external data transmission. The system can audit any enterprise document type, not just contracts.

THE FUNCTIONAL CONTRACT

- **Feature 1: Anonymized Contract Upload & Processing Pipeline**
 - User uploads a legal contract (PDF or DOCX). The system extracts text locally via PDF.js/Mammoth.js, runs spaCy NER to detect and replace sensitive entities (company names, person names, monetary values, NIC numbers, addresses, dates, contacts) with tokens.
 - The anonymized document is displayed side-by-side with the original for reviewer inspection. Entity map is stored locally — no raw sensitive data is transmitted externally
- **Feature 2: Multi-Agent Adversarial Audit Loop with RAG Grounding**
 - Anonymized contract text + RAG-retrieved context (Sri Lankan legal clauses + past Hemas contract annotations + standard templates) is passed to the AUDITOR agent via Groq API (Llama 3.3 70B).
 - AUDITOR produces structured JSON findings.
 - ATTACKER agent receives the same document + AUDITOR findings and returns a gap report.
 - AUDITOR revises under adversarial pressure. Risk score is calculated and evaluated against threshold. Loop iterates if required. Document is flagged HIGH RISK / MODERATE / LOW RISK accordingly
- **Feature 3: Structured Risk Report Generation & Dashboard**
 - Following the audit loop, the system re-maps all tokens to original entity values and generates a structured risk report delivered through the React dashboard and downloadable as PDF.
 - Report includes:
 - overall risk flag status with score
 - executive summary (plain-English narrative for senior decision-maker)
 - severity-rated findings (Critical/High/Medium/Low) with original clause text and recommended actions
 - missing clause checklist
 - audit trail showing AUDITOR vs ATTACKER iterations.

ROI & STRATEGIC ROADMAP

- Projected ROI

Metric	Current State	With ContractIQ	Annual Impact
Review time / contract	3 to 5 days	20–35 minutes	99% time saving
Senior legal capacity	Consumed by routine review	Freed for strategic work	LKR 4.2M+ reclaimed
Review consistency	Reviewer-dependent	Standardized across all SBUs	100% uniform standard

- Strategic Roadmap

Phase	Scope	Enhancement
MVP	COE contract auditing	Core 3 features — anonymization, adversarial loop, risk report
Phase 2	All 6 SBUs	Contract-type-specific templates for pharma, hospital, logistics, manufacturing
Phase 3	On-premise deployment	Replace Groq API with on-premise Llama 3 , zero external data transmission
Phase 4	Enterprise document intelligence	Extend pipeline to audit board papers, regulatory filings, audit reports group-wide

- Governance and Ethical AI Use

ContractIQ ensures secure, ethical AI governance (PDPA 2022) with local spaCy NER for automated data anonymization. Its "Human-in-the-Loop" model requires mandatory human sign-off and provides a traceable audit trail for AI risk findings. Features include configurable risk thresholds and a modular LLM layer to prevent vendor lock-in and allow future on-premise hosting, ensuring a standardized, efficient review across all six Hemas SBUs.